

Zipf

Tags:

Time Limit: 1 s

Memory Limit: 128 MB

Submission: 6

AC: 3

Score: 99.98

Submit

Codes

AI Code Tips

Description

Problem D - Zipf's Law

Harvard linguistics professor George Kingsley Zipf (1902-1950) observed that the frequency of the k th most common word in a text is roughly proportional to $1/k$. He justified his observations in a book titled *Human behavior and the principle of least effort* published in 1949. While Zipf's rationale has largely been discredited, the principle still holds, and others have afforded it a more sound mathematical basis.

Input

Input consists of several test cases. The first line of each case contains a single positive integer n . Several lines of text follow which will contain no more than 10000 words. The text for each case is terminated by a single line containing EndOfText. EndOfText does not appear elsewhere in the input and is not considered a word.

Output

For each test case, output the words which occur n times in the input text, one word per line, lower case, in alphabetical order. If there are no such words in input, output the following line:

```
There is no such word.
```

Leave a blank line between cases.

Samples

input

Copy

2

```
In practice, the difference between theory and practice is always
greater than the difference between theory and practice in theory.
- Anonymous
```

```
Man will occasionally stumble over the truth, but most of the
time he will pick himself up and continue on.
- W. S. L. Churchill
```

EndOfText

output

Copy

```
between
difference
in
will
```

Submit

Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

[--- Maintainer List ---](#)

Server Time: 2025-8-26 15:04:52